

Prabhakaran S

Electronics Hardware Engineer

Contact

Ayyankolly

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Education

BE (EEE) – 2018
(Sri Ramakrishna Engg. College,
Coimbatore)

Dipl. (EEE) – 2015
(Govt. Polytechnic, Ooty)

Key Skills

Programming Language –
C,C++,Python

Micro-controllers – STM32, AVR,
Microchip, ESP32,
8051,Infineon MCU's,FPGA

Protocols – I2C, SPI,
UART,EtherCAT,CAN,
XCP,UDS,LIN,Ethernet

Hardware Design – Design
Spark, Kicad, Altium Designer,
Solid Works

Test & Debug Tools-
Oscilloscope,LogicAnalyzer,Multi
meter,ICDFunctionGenerator,
Power Supply
Soldering&Desoldering Tools,

Others – PCB Assembly, 3D
Printing

Summary

Electronics Hardware Engineer with 7 years of experience in product development, circuit design, embedded software, and PCB design, from board bring-up through design verification testing. Skilled in component lifecycle management, microcontroller programming, and communication protocols.

Experience

September 2023 – Present

Hardware Engineer • Igraph Technologies Pvt. Ltd, Kollapally

Responsibilities included:

- Led hardware design initiatives, specializing in multilayer and high-speed PCB layouts for advanced electronic products, including Battery Management Systems (BMS) and Hardware-in-the-Loop (HIL) systems.
- Designed, maintained, and updated comprehensive PCB documentation, including Gerber files and assembly drawings, supporting efficient fabrication and assembly processes.
- Managed end-to-end product development, from initial concept through to production, ensuring technical excellence and timely project delivery.
- Planned, created, and maintained detailed Bills of Materials (BOM) for all hardware projects, optimizing costs and streamlining component sourcing.

October 2018 – September 2023

Research Engineer • Vingyan Innovations Pvt. Ltd, Chennai

Responsibilities included:

- Spearheaded R&D and embedded development projects, quick turnaround lab prototypes, proof-of-concepts.
- Developed cost-effective methods to automate hardware testing, improving efficiency.
- Develop sample code, reusable software libraries.
- Managed PCB documentation, including fabrication, assembly drawings, and gerber files.

Projects

Falcon – MCU — *Simtestlab* – 2024

- Designed and developed FALCON MCU, a multilayer microcontroller board with a plug-in, plug-out architecture for enhanced flexibility and rapid prototyping.
- Engineered as a universal, multi-purpose platform, supporting multiple communication protocols for broad application versatility.
- Selected optimal electronic components and produced comprehensive schematic drawings to ensure reliable, manufacturable designs.
- Designed intricate Ethernet and EtherCAT routing to ensure robust, high-speed industrial communication.
- Applied advanced EMC (Electromagnetic Compatibility) and EMI (Electromagnetic Interference) practices to minimize noise and guarantee compliance with industry standards.
- Led complete hardware design, multilayer PCB layout, and documentation, ensuring seamless integration with various modules and systems.
- Developed the board with a compact, compatible form factor for easy integration into multiple applications and product enclosures.

HIL – Hardware in Loop — *Simtestlab* – 2025

- Designed and developed a multilayer Hardware-in-the-Loop (HIL) system specifically for Electronic Control Unit (ECU) validation and testing.
- Engineered the HIL platform to handle multiple input/output signals from the ECU, supporting reliable operation at currents up to 16 Amps.
- Created comprehensive schematic diagrams and high-density multilayer PCB layouts to ensure robust connectivity and superior signal integrity.
- Integrated mixed-signal (analog, digital, and power) channels for full-featured ECU simulation and validation scenarios.
- Applied advanced EMC (Electromagnetic Compatibility) and EMI (Electromagnetic Interference) best practices, ensuring noise immunity and compliance with stringent automotive industry standards.
- Documented all hardware designs, PCB layouts, and assembly processes, supporting seamless manufacturing, future upgrades, and scalability.

Pendent – Women safety Device — *Sukra* – 2024

- Designed and developed a women's safety device, integrating hardware and custom firmware for real-time emergency response.
 - Engineered the system to detect panic situations; upon activation, the device captures the user's image and GPS location.
 - Developed embedded firmware to automatically transmit the captured image and location data to pre-assigned guardians and the nearest SOS/emergency services.
 - Ensured fast and reliable communication for critical alerts, enhancing user safety and providing timely assistance during emergencies.
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Hardware Functional Tester – *Simtestlab – 2021 - 2023*

- Developed a multifunction hardware testing device capable of testing various sensors and PCBs – simultaneously and separately
- Implemented a user-friendly interface for configuring and executing hardware tests.
- Facilitated efficient testing of sensor functionality, signal quality, and PCB performance by automating hardware test procedures.
- Collaborated with cross-functional teams to ensure seamless integration into hardware development lifecycle.
- Purpose of this tester is to reduce human intervention during Teckcraft manufacturing & kitting process

Patient Monitoring System — *Client: D4x Technologies – 2022*

- Developed an ESP32-based IoT device for real-time patient monitoring.
- Established UART communication with a pulse oximeter for continuous data collection.
- Designed the system to collect patient vital signs, such as heart rate and oxygen saturation levels.
- Implemented cloud connectivity to securely upload patient data to a cloud database.
- Enabled remote monitoring of patients' health status from anywhere with internet access.

RC-Truck — *Hobby Project – 2024*

- Designed and 3D-printed a 1:18 scale RC truck, closely replicating the aesthetics and mechanical systems of a real truck.
- Developed a custom multi-function, multi-channel transmitter and receiver system, enabling robust remote control with a range of up to 1 km.
- Integrated a live video transmission system to deliver real-time driver's-view feedback for an immersive driving experience.
- Created a separate dashboard replicating authentic truck indicators and lighting systems, enhancing realism with functional displays and controls.
- Engineered and fabricated the entire truck chassis, axle, and driving system, providing accurate steering, suspension, and drive characteristics.

Designed both the hardware and firmware to simulate the genuine feel and features of operating a full-sized truck.

References
